installer. Both the packages are available in Python 2 base and Python 3 base. Installation steps for different operating systems are given below.

Installation in Windows: Download the *exe* installer for Anaconda or Miniconda. Run the file to get it installed. To open it in a terminal, go to Start button and click on Run, then open the command prompt (cmd).

Installation in Linux: Download the *bash* installer for Anaconda or Miniconda. Type the following command in a terminal to install 64-bit Python 3 based Miniconda:

```
bash Miniconda3-latest-Linux-x86_64.sh
```

…where *Miniconda3-latest-Linux-x86_64* is the name of the file you have downloaded.

Once the installation is completed, close the terminal before using Conda. This is done to make sure that the changes made are saved.

Installation in macOS: Anaconda provides a command-line installer and GUI installer for macOS users. If you choose the GUI installer, double click on the *.pkg* file downloaded, and follow the instructions to get it installed in your system. The GUI installer may take more time. So if you are comfortable with the command-line installer, go for it.

If you download the command-line installer, follow the same procedure as for installation in Linux. You must remember that even if you are not using the *bash* shell, you must include the *bash* command for installation.

Miniconda installation is the same as the command-line installation of Anaconda.

Conda without Anaconda or Miniconda: Conda can also be installed using *pip*, with the following command:

```
pip install conda
```

This command will install Conda without Anaconda or Miniconda. This method can be adopted easily in Linux. But it is difficult to install *pip* in Windows. *pip* comes along with Python 2.7.9 and above.

To update Conda, type the following command in the terminal:

```
conda update conda
```

In Windows, Conda can be uninstalled by following the steps given below:

1. Go to *Control Panel*.
2. Select *Add or Remove Program*.
3. Select *Python 3.4(Miniconda)* and uninstall it.

In Linux and macOS, use the following command to uninstall the Miniconda directory:

```
rm -rf ~/miniconda
```

The Miniconda install directory will now be deleted. But you may still be able to access the packages. To delete Miniconda completely from the system, edit *~/.bash_profile* and remove the Miniconda directory from the *PATH* variable. You will no longer have access to Conda packages.

To verify the installation of Conda, type:

```
conda list
```

This command will display the installed packages in the terminal if the Conda installation is successful. Otherwise, a message that 'Conda is not recognised as an internal or external command, operable program or batch file' will be displayed on the screen.

Why do we need Conda?

Continuum Analytics has developed Conda with a view to supporting data analysis and scientific computing applications. Scientific applications handle huge amounts of data. And a variety of packages may be needed to process such large volumes of data. Conda uses the rich repository of Anaconda, which contains almost all the necessary packages for scientific programming. It is an alternative to the above mentioned package managers.

Another important use of Conda is in creating a virtual environment. You can use *virtual box* to create a virtual environment. In that case, you are using a separate platform which needs separate resources. But a Conda generated virtual environment doesn't need a separate platform. It is very easy to get into the environment and to get out of it. You may be familiar with *virtualenv*, which is a tool similar to Conda. A comparison of both can be found in the link *https://conda.io/docs/_downloads/conda-pip-virtualenv-translator.html*

Conda allows you to install different versions of the same package on the same machine but in different environments. Suppose you need Matplotlib 1.4 to run an application and Matplotlib 1.5 is needed for another application. A single environment cannot accommodate both these versions at the same time. Since Matplotlib 1.5 is the upgraded version of Matplotlib 1.4, you can use Conda to create different environments and install the different versions in them; you can then run the applications in their respective environments without any trouble.

Everything related to an environment is localised. If you install a package in a root directory, its dependencies and related information will be dispersed in different directories in the system and hence deletion may not be possible by using a single command. In case of virtual environments, everything related to an environment will be stored in a single directory. So once the environment is deleted, everything related to that will also be automatically deleted.